Mitchell A. Goldman

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Education

B.S. in Mechanical Engineering, Entrepreneurship and Innovation Minor **GPA 3.87/4.00**

The Pennsylvania State University, *University Park, PA* Graduation Date: May 2027

Awards: Dean's List, The President Walker Award, Outdoor Adventures Outstanding Staff Member

Engineering Experience

Boeing Intern Program **May 2026 - Present**

Enterprise Certification Data Safety & Airworthiness Engineering *Everett, WA*

- Built a capability kit SharePoint and Power Apps platform to reduce effort in locating and maintaining critical technical content for enterprise certification processes.
- Developed a dynamic visual document tree to map Command Media relationships, ownership, and hierarchy, improving traceability and accessibility of controlled information across the organization.
- Leveraged Codex CLI and internal APIs to identify and resolve misalignments between project timelines, requirements, and DevOps workflows.

Summers By Design Galapagos Study Abroad **May 2025**

Penn State Education Abroad *Galápagos & Quito, Ecuador*

- Investigated sustainability challenges, user needs, and UN SDG-aligned opportunities for local tourism businesses through bilingual interviews, then refined 20+ design concepts into Clima Key, a scalable A/C efficiency solution projected to reduce electricity use in Ecuador by 5%.

Design Team Director **April 2024 - Present**

The Penn State American Society of Mechanical Engineers (PSU ASME) *State College, PA*

- Directed operations for a 250+ member multidisciplinary engineering organization, overseeing 9 concurrent projects and mentoring project leads in innovative problem solving, team management, and project delivery.
- Led a 20-member engineering team to a 3rd-place national finish at the ASME EFx Student Design Challenge by developing a sustainability-focused waste collection and sorting robot.
- Executed a 200+ attendee intercollegiate combat robotics competition involving 13 teams across 9 organizations, coordinating operations, safety, sponsorships, and stakeholder communication.
- Constructed and delivered technical workshops in CAD, manufacturing, generative design, Arduino, MATLAB, and portfolio building to support project work, strengthen innovation, and enhance professional skills.

Research Assistant **February 2024 - May 2024**

Penn State Biological and Robotic Intelligent Fluid Locomotion (BioRob-InFL) Lab *State College, PA*

- Validated digital twin simulations through 60+ real-world flight trials, confirming neural network and PID algorithms that enable reliable inverted drone perching at 3.5 m/s.
- Publication: B. Habas, A. Brown, D. Lee, **M. Goldman**, B. Cheng, "From Ceilings to Walls: Universal Dynamic Perching of Small Aerial Robots on Surfaces with Variable Orientations" - IEEE ICRA, 2025.

Leadership and Philanthropy

Food Recovery Network **January 2025 - Present**

Penn State FRN *State College, PA*

- Collect 50+ leftover meals in the Penn State dining halls weekly and donate them to local food banks.

Outdoor Adventures Supervisor **August 2023 - Present**

Penn State Outdoor Adventures *State College, PA*

- Train and supervise 26 climbing wall staff in risk management, customer service, and professionalism.
- Instruct climbing technique lessons twice a week and facilitate high-ropes programs once a week for 40+ participants, fostering a safe, educational, and engaging environment for individual and team growth.

Skills

- Technical: nTop, Autodesk Fusion, Onshape, SolidWorks, User-centered Design, Design for Manufacturing, Generative Design, C++, Python, MATLAB, Arduino, Additive manufacturing, Microsoft 365
- Certifications: Autodesk Certified Associate in CAD for Mechanical Design, Red Cross CPR, First Aid, and AED